

Opportunity & Objectives

The measurable targets this design is held to

Objectives that cannot be measured cannot be failed, and objectives that cannot be failed are not objectives. Every target below is stated as a number, tied to the analysis that produces it, and reported against in this submission — including where the result is modest.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
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Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. From ranked problems to measurable targets

The problem definition ranks thermal discomfort as the critical constraint. The objectives below follow from that ranking rather than from a general list of good intentions, and each one names the analysis that verifies it.

Objective	Target	Achieved	Verified by
Raise comfortable daylight hours	$\geq 60\%$	64.6%	8,760-hour comfort model
Provide a continuously shaded primary route	$\geq 85\%$ of daylight hours	87.3%	Ray-traced section across the year
Minimise hours with no shade anywhere on the route	Minimise	52 h, from 330 h	Plan-form sweep, src/config.py
Reduce peak heat index	$\geq 5\text{ }^{\circ}\text{C}$	7.13 $^{\circ}\text{C}$ mean	Heat-index model, shaded vs exposed
Deliver within budget	$\leq$ AED 35 M	AED 27.0 M (77%)	Capital cost plan, src/costing.py
Meet the brief's minimum programme	All items	20 facilities placed	Facilities map, src/plan.py

Each target is regenerated and re-checked on every pipeline run.

2. An objective the scheme deliberately does not maximise

Site-wide mean shade is 34.1%, and no target was set for it. Maximising average shade across 15,000 m² would spread the budget thin and produce a park that is uniformly mediocre. The scheme instead concentrates its shade into one continuous, genuinely excellent route. That is a design position, and it is stated here so the modest site-wide figure is read as intent rather than as failure.

3. Objectives beyond thermal comfort

- **Universal accessibility.** The site is level, the whole park is step-free, and the only change of level is ramped.
- **Day and night activation.** The convex face carries the evening uses; the crescent is lit from within.
- **Operational viability.** Commercial floor is placed to support running costs, and every facility is serviceable without a vehicle entering the walk.
- **Water discipline.** Open water is limited to a shaded channel; irrigation is subsurface drip on treated effluent.
- **Reproducibility.** Every claim in this submission can be regenerated from source data with one command.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.